Abhijit Shidlagatta

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PERSONAL PROFILE

Computer Science graduate (2:1) specialising in real-time graphics and gameplay systems in C++, Unreal Engine and HLSL, with hands-on experience in game development, multithreading, procedural generation and custom GPU shaders. Graded 86.2% for my final-year dissertation and 78% in Computer Graphics.

EDUCATION

City St George's, University of London *Sep 2023 - Jun 2026*
BSc (Hons) Computer Science **Upper Second-Class Honours (2:1)**

- **Relevant Modules:** Computer Graphics (78%), Advanced Games Technology (69%), Games Technology, Programming in C++, Object-Oriented Analysis and Design.

Tiffin School *Completed 2021*
A-Levels: Mathematics (A*), Physics (B), Economics (B) | **GCSEs:** 12 subjects including Maths (8) & English

TECHNICAL PROJECTS

Real-Time Procedural Ocean Renderer | *C++, Unreal Engine 5, HLSL* *2026*

- Implemented Tessendorf's FFT ocean simulation from scratch as HLSL compute shaders, building a fully custom GPU spectral wave pipeline instead of using Unreal's built-in Water plugin.
- Wrote a radix-2 Cooley-Tukey FFT as four compute passes (spectrum initialisation, time evolution, butterfly, inversion), running three wave cascades every frame through Unreal's Render Dependency Graph (RDG).
- Built the water colour on Unreal's Single Layer Water model, using Pope & Fry's measured pure-water absorption for open ocean and hand-tuned Tropical/Murky presets, switchable at runtime with one call.

Dissertation: Procedural Destructible Voxel Environment (Grade: 86.2%) | *C++, Unreal Engine 5 2025-2026*

- Built a real-time procedural dungeon generator in C++ and UE5, balancing dynamic voxel destructibility with structured room layouts.
- Validated room placement with a buffered AABB range query on the voxel octree before carving, so new rooms never overlap existing ones, falling back to a connecting tunnel when the space is taken.
- Moved the $O(n^3)$ voxel carving loop onto background worker threads via Unreal's AsyncTask API with voxel write locks, cutting the 1% low frame time during room generation from 45.2 ms to 17.1 ms.

University Games Projects: OpenGL Racing Game & C++ Souls-like | *C++, OpenGL, GLSL* *2025-2026*

- **Computer Graphics:** off-road racing game on a Catmull-Rom spline track, built on the module's OpenGL framework, with bloom, shadow mapping, reflections, a water shader, instanced foliage, motion blur and frustum culling.
- **Advanced Games Technology:** "Souls-like" action RPG in C++ on the module's teaching engine, with an invulnerable dodge-roll, state-machine enemy AI, a boss with timed damage windows and a soul-based upgrade economy.

WebGL ASCII Render Pipeline | *GLSL, WebGL, Three.js* *2026*

- Built a GLSL fragment-shader pipeline that converts images, video and webcam feeds into ASCII art live.
- Implemented GPU image processing, including a real-time 3x3 Sobel operator for directional edge detection and Bayer matrix ordered dithering.

Database-Linked Calendar System (Team Project, 72.5%) | *Java, SQL* *2024-2025*

- Integrated a Java back-end with an SQL database.
- Handled version control and merging code in an Agile student team to meet the business requirements.

Indie Game Jams (GMTK) | *Unity, Godot, C#* *2026*

- Designed and coded complete games in 48-hour jams, handling gameplay code, debugging and art integration.
- Designed and drew every sprite in *Sign of the Times* (GMTK Game Jam 2026, Unity), and for a PSX-style horror jam game hand-animated the first-person item assets in Krita and modelled 3D assets in Blender.

TECHNICAL SKILLS

Programming: C++, GLSL/HLSL, C#, Java, Python, JavaScript/TypeScript
Tools & Technologies: Git/GitHub, Unreal Engine, Unity, WebGL, OpenGL, Three.js, Blender, Krita
Engineering & Rendering: Multithreading, Memory Optimisation, Procedural Generation, Shaders, Graphics Pipelines
Art & Pipelines: Technical Art Pipelines, 3D Modelling, Digital Illustration, Level Design